

These equations were proposed by Rikitake (1958) as a model for the self-generation of the Earth's magnetic field by large current-carrying eddies in the core. Computer experiments show that the model exhibits chaotic solutions for some parameter values. These solutions are loosely analogous to the irregular reversals of the Earth's magnetic field inferred from geological data. See Cox (1982) for the geophysical background.

9.3 Chaos on a Strange Attractor

9.3.1 (Quasiperiodicity $\neq$ chaos) The trajectories of the quasiperiodic system $\dot{\theta}_1 = \omega_1$, $\dot{\theta}_2 = \omega_2$, (ω_1/ω_2 irrational) are not periodic.

- Why isn't this system considered chaotic?
- Without using a computer, find the largest Liapunov exponent for the system.

(Numerical experiments) For each of the values of r given below, use a computer to explore the dynamics of the Lorenz system, assuming $\sigma = 10$ and $b = 8/3$ as usual. In each case, plot $x(t)$, $y(t)$, and x vs. z . You should investigate the consequences of choosing different initial conditions and lengths of integration. Also, in some cases you may want to ignore the transient behavior, and plot only the sustained long-term behavior.

9.3.2 $r = 10$

9.3.3 $r = 22$ (transient chaos)

9.3.4 $r = 24.5$

9.3.5 $r = 100$ (surprise)

(chaos and stable point co-exist)

9.3.6 $r = 126.52$

9.3.7 $r = 400$

9.3.8 (Practice with the definition of an attractor) Consider the following familiar system in polar coordinates: $\dot{r} = r(1 - r^2)$, $\dot{\theta} = 1$. Let D be the disk $x^2 + y^2 \leq 1$.

- Is D an invariant set?
- Does D attract an open set of initial conditions?
- Is D an attractor? If not, why not? If so, find its basin of attraction.
- Repeat part (c) for the circle $x^2 + y^2 = 1$.

9.3.9 (Exponential divergence) Using numerical integration of two nearby trajectories, estimate the largest Liapunov exponent for the Lorenz system, assuming that the parameters have their standard values $r = 28$, $\sigma = 10$, $b = 8/3$.

9.3.10 (Time horizon) To illustrate the "time horizon" after which prediction becomes impossible, numerically integrate the Lorenz equations for $r = 28$, $\sigma = 10$, $b = 8/3$. Start two trajectories from nearby initial conditions, and plot $x(t)$ for both of them on the same graph.

9.4 Lorenz Map

9.4.1 (Computer work) Using numerical integration, compute the Lorenz map for $r = 28$, $\sigma = 10$, $b = 8/3$.

9.4.2 (Tent map, as model of Lorenz map) Consider the map

$$x_{n+1} = \begin{cases} 2x_n, & 0 \leq x_n \leq \frac{1}{2} \\ 2 - 2x_n, & \frac{1}{2} \leq x_n \leq 1 \end{cases}$$

as a simple analytical model of the Lorenz map.

- Why is it called the “tent map”?
- Find all the fixed points, and classify their stability.
- Show that the map has a period-2 orbit. Is it stable or unstable?
- Can you find any period-3 points? How about period-4? If so, are the corresponding periodic orbits stable or unstable?

9.5 Exploring Parameter Space

(Numerical experiments) For each of the values of r given below, use a computer to explore the dynamics of the Lorenz system, assuming $\sigma = 10$ and $b = 8/3$ as usual. In each case, plot $x(t)$, $y(t)$, and x vs. z .

9.5.1 $r = 166.3$ (intermittent chaos)

9.5.2 $r = 212$ (noisy periodicity)

9.5.3 the interval $145 < r < 166$ (period-doubling)

9.5.4 (Hysteresis between a fixed point and a strange attractor) Consider the Lorenz equations with $\sigma = 10$ and $b = 8/3$. Suppose that we slowly “turn the r knob” up and down. Specifically, let $r = 24.4 + \sin \omega t$, where ω is small compared to typical orbital frequencies on the attractor. Numerically integrate the equations, and plot the solutions in whatever way seems most revealing. You should see a striking hysteresis effect between an equilibrium and a chaotic state.

9.5.5 (Lorenz equations for large r) Consider the Lorenz equations in the limit $r \rightarrow \infty$. By taking the limit in a certain way, all the dissipative terms in the equations can be removed (Robbins 1979, Sparrow 1982).

- Let $\varepsilon = r^{-1/2}$, so that $r \rightarrow \infty$ corresponds to $\varepsilon \rightarrow 0$. Find a change of variables involving ε such that as $\varepsilon \rightarrow 0$, the equations become

$$\begin{aligned} X' &= Y \\ Y' &= -XZ \\ Z' &= XY. \end{aligned}$$

- Find two conserved quantities (i.e., constants of the motion) for the new system.
- Show that the new system is volume-preserving (i.e., the volume of an arbitrary blob of “phase fluid” is conserved by the time-evolution of the system, even though the shape of the blob may change dramatically.)
- Explain physically why the Lorenz equations might be expected to show some conservative features in the limit $r \rightarrow \infty$.

- e) Solve the system in part (a) numerically. What is the long-term behavior? Does it agree with the behavior seen in the Lorenz equations for large r ?

9.5.6 (Transient chaos) Example 9.5.1 shows that the Lorenz system can exhibit transient chaos for $r = 21$, $\sigma = 10$, $b = \frac{8}{3}$. However, not all trajectories behave this way. Using numerical integration, find three different initial conditions for which there *is* transient chaos, and three others for which there *isn't*. Give a rule of thumb which predicts whether an initial condition will lead to transient chaos or not.

9.6 Using Chaos to Send Secret Messages

9.6.1 (Exponentially fast synchronization) The Liapunov function of Example 9.6.1 shows that the synchronization error $\mathbf{e}(t)$ tends to zero as $t \rightarrow \infty$, but it does not provide information about the rate of convergence. Sharpen the argument to show that the synchronization error $\mathbf{e}(t)$ decays exponentially fast.

- Prove that $V = \frac{1}{2}e_2^2 + 2e_3^2$ decays exponentially fast, by showing $\dot{V} \leq -kV$, for some constant $k > 0$ to be determined.
- Show that part (a) implies that $e_2(t), e_3(t) \rightarrow 0$ exponentially fast.
- Finally show that $e_1(t) \rightarrow 0$ exponentially fast.

9.6.2 (Pecora and Carroll's approach) In the pioneering work of Pecora and Carroll (1990), one of the receiver variables is simply set *equal to* the corresponding transmitter variable. For instance, if $x(t)$ is used as the transmitter drive signal, then the receiver equations are

$$\begin{aligned} \dot{x}(t) &= x(t) \\ \dot{y}_r &= rx(t) - y_r - x(t)z_r \\ \dot{z}_r &= x(t)y_r - bz_r \end{aligned}$$

where the first equation is *not* a differential equation. Their numerical simulations and a heuristic argument suggested that $y_r(t) \rightarrow y(t)$ and $z_r(t) \rightarrow z(t)$ as $t \rightarrow \infty$, even if there were differences in the initial conditions.

Here is a simple proof of that result, due to He and Vaidya (1992).

- Show that the error dynamics are

$$\begin{aligned} e_1 &\equiv 0 \\ \dot{e}_2 &= -e_2 - x(t)e_3 \\ \dot{e}_3 &= x(t)e_2 - be_3 \end{aligned}$$

where $e_1 = x - x_r$, $e_2 = y - y_r$, and $e_3 = z - z_r$.

- Show that $V = e_2^2 + e_3^2$ is a Liapunov function.
- What do you conclude?

9.6.3 (Computer experiments on synchronized chaos) Let x, y, z be governed by the Lorenz equations with $r = 60$, $\sigma = 10$, $b = 8/3$. Let x_r, y_r, z_r be governed by the system in Exercise 9.6.2. Choose different initial conditions for y and y_r , and similarly for z and z_r , and then start integrating numerically.

- Plot $y(t)$ and $y_r(t)$ on the same graph. With any luck, the two time series should eventually merge, even though both are chaotic.
- Plot the (y, z) projection of both trajectories.

9.6.4 (Some drives don't work) Suppose $z(t)$ were the drive signal in Exercise 9.6.2, instead of $x(t)$. In other words, we replace z_r by $z(t)$ everywhere in the receiver equations, and watch how x_r and y_r evolve.

- Show numerically that the receiver does *not* synchronize in this case.
- What if $y(t)$ were the drive?

9.6.5 (Masking) In their signal-masking approach, Cuomo and Oppenheim (1992, 1993) use the following receiver dynamics:

$$\begin{aligned}\dot{x}_r &= \sigma(y_r - x_r) \\ \dot{y}_r &= rs(t) - y_r - s(t)z_r \\ \dot{z}_r &= s(t)y_r - bz_r\end{aligned}$$

where $s(t) = x(t) + m(t)$, and $m(t)$ is the low-power message added to the much stronger chaotic mask $x(t)$. If the receiver has synchronized with the drive, then $x_r(t) \approx x(t)$ and so $m(t)$ may be recovered as $\hat{m}(t) = s(t) - x_r(t)$. Test this approach numerically, using a sine wave for $m(t)$. How close is the estimate $\hat{m}(t)$ to the actual message $m(t)$? How does the error depend on the frequency of the sine wave?

9.6.6 (Lorenz circuit) Derive the circuit equations for the transmitter circuit shown in Figures 9.6.1.

ONE-DIMENSIONAL MAPS

10.0 Introduction

This chapter deals with a new class of dynamical systems in which time is *discrete*, rather than continuous. These systems are known variously as difference equations, recursion relations, iterated maps, or simply *maps*.

For instance, suppose you repeatedly press the cosine button on your calculator, starting from some number x_0 . Then the successive readouts are $x_1 = \cos x_0$, $x_2 = \cos x_1$, and so on. Set your calculator to radian mode and try it. Can you explain the surprising result that emerges after many iterations?

The rule $x_{n+1} = \cos x_n$ is an example of a ***one-dimensional map***, so-called because the points x_n belong to the one-dimensional space of real numbers. The sequence $x_0, x_1, x_2, \dots$ is called the ***orbit*** starting from x_0 .

Maps arise in various ways:

1. *As tools for analyzing differential equations.* We have already encountered maps in this role. For instance, Poincaré maps allowed us to prove the existence of a periodic solution for the driven pendulum and Josephson junction (Section 8.5), and to analyze the stability of periodic solutions in general (Section 8.7). The Lorenz map (Section 9.4) provided strong evidence that the Lorenz attractor is truly strange, and is not just a long-period limit cycle.
2. *As models of natural phenomena.* In some scientific contexts it is natural to regard time as discrete. This is the case in digital electronics, in parts of economics and finance theory, in impulsively driven mechanical systems, and in the study of certain animal populations where successive generations do not overlap.
3. *As simple examples of chaos.* Maps are interesting to study in their own right, as mathematical laboratories for chaos. Indeed, maps are capable

of much wilder behavior than differential equations because the points x_n hop along their orbits rather than flow continuously (Figure 10.0.1).



Figure 10.0.1

The study of maps is still in its infancy, but exciting progress has been made in the last few decades, thanks to the growing availability of calculators, then computers, and now computer graphics. Maps are easy and fast to simulate on digital computers where time is *inherently* discrete. Such computer experiments have revealed a number of unexpected and beautiful patterns, which in turn have stimulated new theoretical developments. Most surprisingly, maps have generated a number of successful predictions about the routes to chaos in semiconductors, convecting fluids, heart cells, lasers, and chemical oscillators.

We discuss some of the properties of maps and the techniques for analyzing them in Sections 10.1–10.5. The emphasis is on period-doubling and chaos in the logistic map. Section 10.6 introduces the amazing idea of universality, and summarizes experimental tests of the theory. Section 10.7 is an attempt to convey the basic ideas of Feigenbaum’s renormalization technique.

As usual, our approach will be intuitive. For rigorous treatments of one-dimensional maps, see Devaney (1989) and Collet and Eckmann (1980).

10.1 Fixed Points and Cobwebs

In this section we develop some tools for analyzing one-dimensional maps of the form $x_{n+1} = f(x_n)$, where f is a smooth function from the real line to itself.

A Pedantic Point

When we say “map,” do we mean the function f or the difference equation $x_{n+1} = f(x_n)$? Following common usage, we’ll call *both* of them maps. If you’re disturbed by this, you must be a pure mathematician . . . or should consider becoming one!

Fixed Points and Linear Stability

Suppose x^* satisfies $f(x^*) = x^*$. Then x^* is a **fixed point**, for if $x_n = x^*$ then $x_{n+1} = f(x_n) = f(x^*) = x^*$; hence the orbit remains at x^* for all future iterations.

To determine the stability of x^* , we consider a nearby orbit $x_n = x^* + \eta_n$ and ask whether the orbit is attracted to or repelled from x^* . That is, does the deviation η_n grow or decay as n increases? Substitution yields